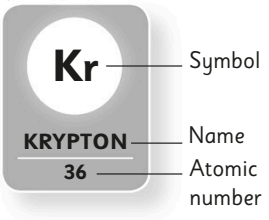
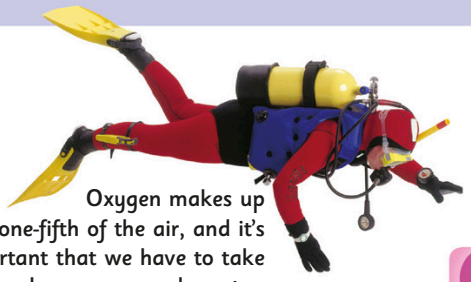
Every element has a name—a symbol made of usually one or two letters—and an atomic number. The atomic number is the number of protons in one atom of the element.





Oxygen makes up about one-fifth of the air, and it's so important that we have to take it with us when we are underwater.

10	11	12	13	14	15	16	17	18
								He HELIUM 2
			B BORON 5	C CARBON 6	N NITROGEN 7	O OXYGEN 8	F FLUORINE 9	Ne NEON 10
			Al ALUMINUM 13	Si SILICON 14	P PHOSPHORUS 15	S SULFUR 16	Cl CHLORINE 17	Ar ARGON 18
Ni NICKEL 28	Cu COPPER 29	Zn ZINC 30	Ga GALLIUM 31	Ge GERMANIUM 32	As ARSENIC 33	Se SELENIUM 34	Br BROMINE 35	Kr KRYPTON 36
Pd PALLADIUM 46	Ag SILVER 47	Cd CADMIUM 48	In INDIUM 49	Sn TIN 50	Sb ANTIMONY 51	Te TELLURIUM 52	I IODINE 53	Xe XENON 54
Pt PLATINUM 78	Au GOLD 79	Hg MERCURY 80	Tl THALLIUM 81	Pb LEAD 82	Bi BISMUTH 83	Po POLONIUM 84	At ASTATINE 85	Rn RADON 86
Ds DARMSTADTIUM 110	Rg ROENTGENIUM 111							

Eu EUROPIUM 63	Gd GADOLINIUM 64	Tb TERBIUM 65	Dy DYSPROSIUM 66	Ho HOLMIUM 67	Er ERBIUM 68	Tm THULIUM 69	Yb YTTERBIUM 70	Lu LUTETIUM 71
Am AMERICIUM 95	Cm CURIUM 96	Bk BERKELIUM 97	Cf CALIFORNIUM 98	Es EINSTEINIUM 99	Fm FERMIUM 100	Md MENDELEVIUM 101	No NOBELIUM 102	Lr LAWRENCIUM 103

KEY:

- **Alkali metals:** These silvery metals are very reactive.
- **Alkaline-earth metals:** These shiny, silvery white metals are reactive.
- **Transition metals:** Many are strong and have high boiling and melting points.
- **Lanthanides:** Many are soft, shiny, and silvery white metals.
- **Actinides:** These are radioactive heavy elements.
- **Poor metals:** These are softer, weaker metals.
- **Non-metals:** Most are gases at room temperature and easily snap as solids.
- **Halogens:** These non-metals are highly reactive and harmful.
- **Noble gases:** These non-metals are the least reactive of all the elements.

Turn and learn

Elements: **pp. 64-65**
Electricity: **pp. 76-77**

Useful elements

We use elements to make all sorts of useful or decorative objects.



Gold is a precious metal. It is used to make jewelry.



Copper is a metal that conducts electricity well. It is used in electrical wires.



Silicon is a non-metal used to make the chips that power computers.



Carbon fibers are strong but light, so they are used for tennis rackets.



Iron is a strong, silvery metal. It is magnetic and has many uses.



Aluminum is a soft, shiny metal. It is used to make soda cans.



Sulfur is a yellow non-metal used to harden rubber to make tires.



Titanium is a very strong, light metal used in aeroplane bodies and space rockets.



Helium is a gas used in balloons because it is less dense than air.



Chlorine is a yellow-green gas, used in bleach and to make some plastics.



Mercury is a liquid metal used in dental fillings and is the gas inside fluorescent lightbulbs.

Technetium (Tc).